Learning Low-Rank Connectivity with Dale's Law Constraints

Overview

This procedure recovers a low-rank estimate of a neural connectivity matrix J from time-series data, respecting Dale's law (neurons are either excitatory or inhibitory). It proceeds in two stages: (1) regress J from data, then (2) decompose J into low-rank factors via structured non-negative matrix factorization (NMF).

Stage 1: Learning J from Data

We assume a linear dynamical system over observed neural activity $y_t \in \mathbb{R}^N$:

$$y_{t+1} = J\, y_t$$

Given recorded trajectories, we stack consecutive observations:

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_{T-1} \end{pmatrix}, \quad Y' = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ y_T \end{pmatrix}$$

and solve the constrained optimization:

$$\min_J \; \|Y'^T - J\, Y^T\|_F$$

subject to Dale's law constraints on the columns of J:

$$J_{:,\,j} \geq 0 \quad \text{if neuron } j \text{ is excitatory,} \qquad J_{:,\,j} \leq 0 \quad \text{if neuron } j \text{ is inhibitory}$$

This is a convex problem solved via CVXPY with the MOSEK solver. For networks with $N > 100$, each row of J is solved independently to reduce memory cost.

Stage 2: Structured NMF Decomposition

The goal is to find low-rank factors $U \in \mathbb{R}^{N \times r}$ and $V \in \mathbb{R}^{r \times N}$ such that $J \approx UV$, where $r =$

$r_e + r_i$.

Step 2a: Split by Cell Type

Partition the rows of J by cell identity:

$$J_E = J[\text{excitatory rows},\ :] \in \mathbb{R}^{N_e \times N}, \quad J_I = J[\text{inhibitory rows},\ :] \in \mathbb{R}^{N_i \times N}$$

Step 2b: Take Absolute Values

Since NMF requires non-negative inputs:

$$\tilde{J}_E = |J_E|, \quad \tilde{J}_I = |J_I|$$

Step 2c: Run NMF

Apply NMF separately to each block:

$$\tilde{J}_E \approx W_E\, H_E, \quad W_E \in \mathbb{R}^{N_e \times r_e},\ H_E \in \mathbb{R}^{r_e \times N}$$

$$\tilde{J}_I \approx W_I\, H_I, \quad W_I \in \mathbb{R}^{N_i \times r_i},\ H_I \in \mathbb{R}^{r_i \times N}$$

with all factors non-negative. This is repeated over 10 random initializations.

Step 2d: Assemble and Restore Signs

Construct the full factorization:

$$U = \begin{pmatrix} W_E & 0 \\ 0 & W_I \end{pmatrix} \in \mathbb{R}^{N \times r}, \quad V = \begin{pmatrix} H_E \\ H_I \end{pmatrix} \in \mathbb{R}^{r \times N}$$

Restore Dale's law by negating the inhibitory columns of V:

$$V_{:,\,j} \leftarrow -V_{:,\,j} \quad \text{for all inhibitory neurons } j$$

The recovered matrix is then $J_{\text{rec}} = UV$, and the best run is selected by minimizing the relative Frobenius error:

$$\epsilon = \frac{\|UV - J\|_F}{\|J\|_F}$$

Stage 3: Initializing a State-Space Model

The factors from Stage 2 are used to initialize the parameters of a linear dynamical system:

$$x_{t+1} = A\,x_t + \text{noise}, \quad y_t = C\,x_t + d + \text{noise}$$

where:

- The **emission matrix** is set to $C = U$, mapping latent states $x_t \in \mathbb{R}^r$ to observations $y_t \in \mathbb{R}^N$.
- The **dynamics matrix** is set to $A = VU \in \mathbb{R}^{r \times r}$, governing latent-state evolution.
- The **bias** d is set to the mean of the data.
- **Noise covariances** for both dynamics and emissions are initialized from the residual variance.

This provides a structured, biologically informed initialization for subsequent EM fitting.

Variant: Shared Row-Space NMF

An alternative constrains $H_E = H_I = H$, so that excitatory and inhibitory populations share the same row space:

$$\tilde{J}_E \approx W_E\,H, \quad \tilde{J}_I \approx W_I\,H$$

This is solved via projected gradient descent, alternating updates of (W_E, W_I) and H with non-negativity projections.